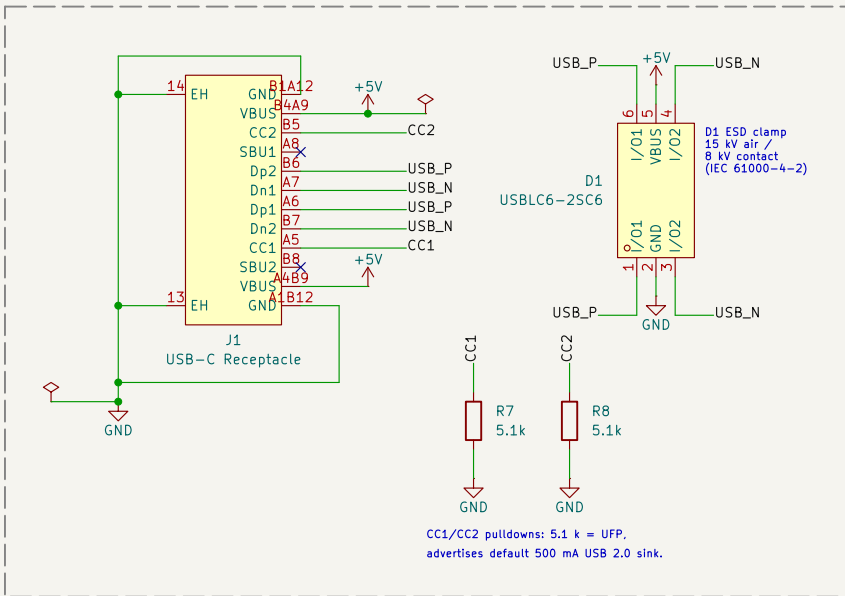
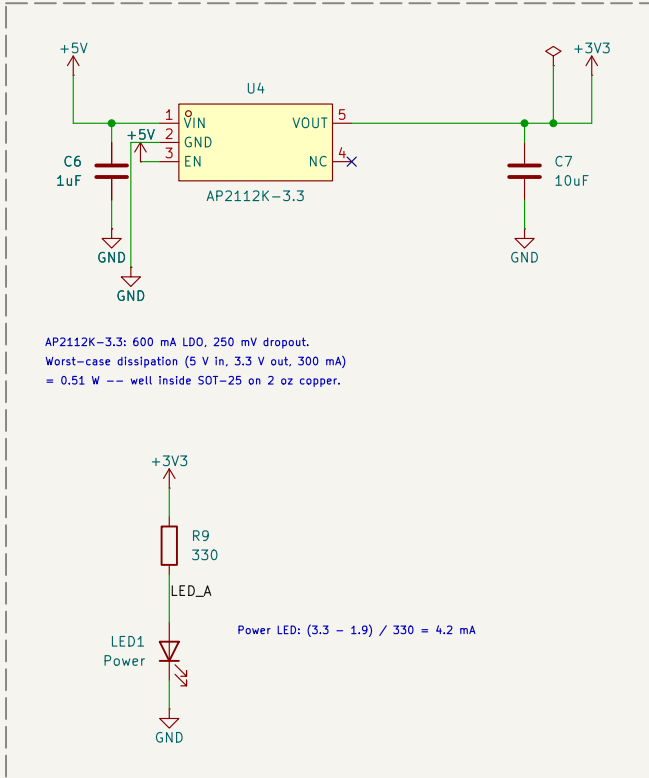


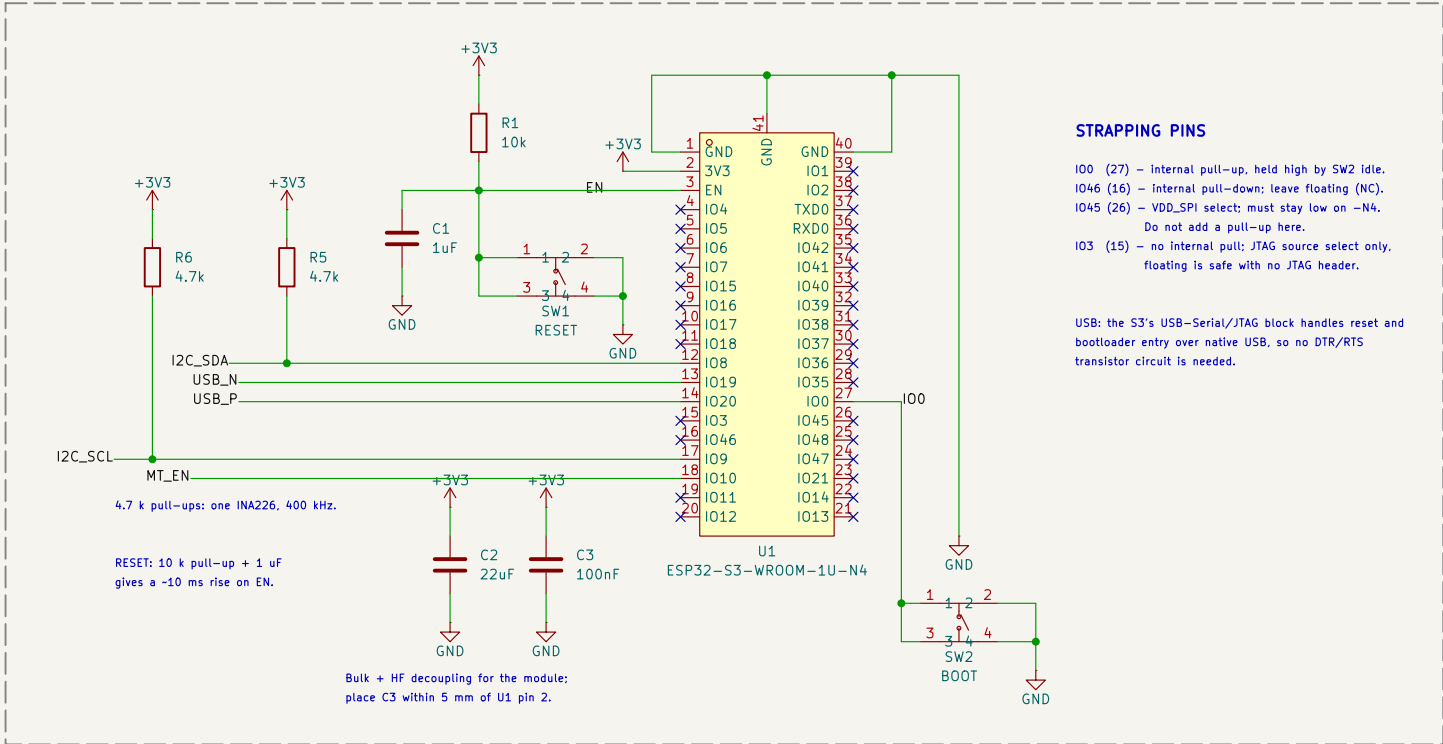
USB-C INPUT & ESD PROTECTION



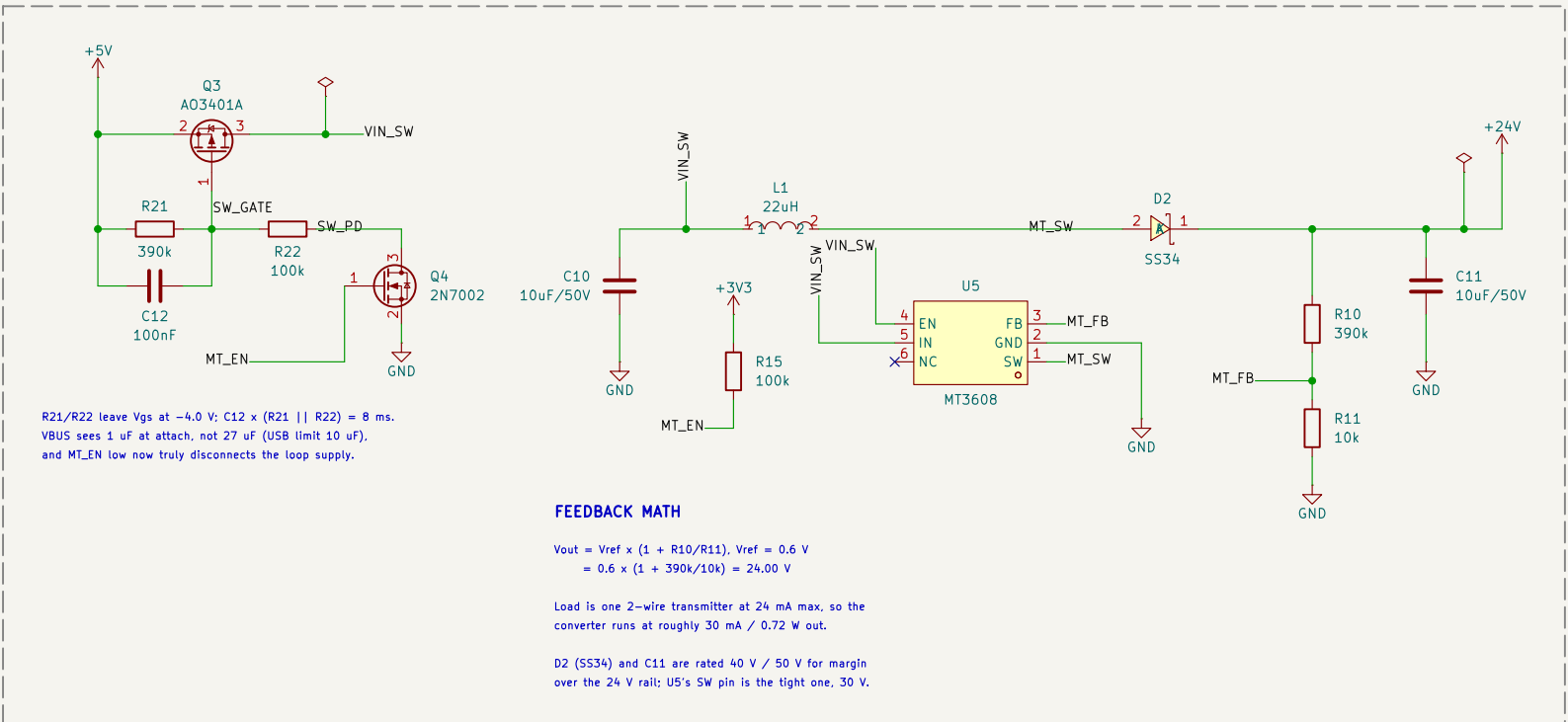
3.3 V REGULATOR & POWER LED



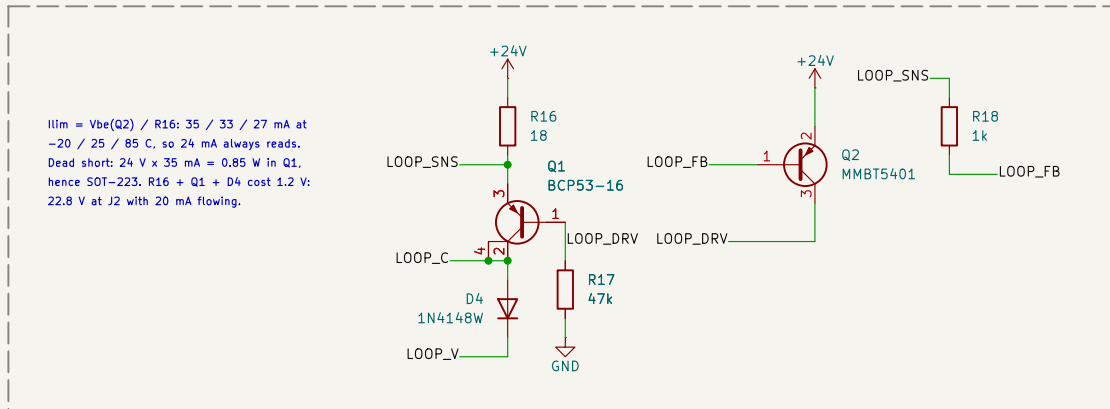
ESP32-S3 MODULE - reset, boot, I2C pull-ups, decoupling



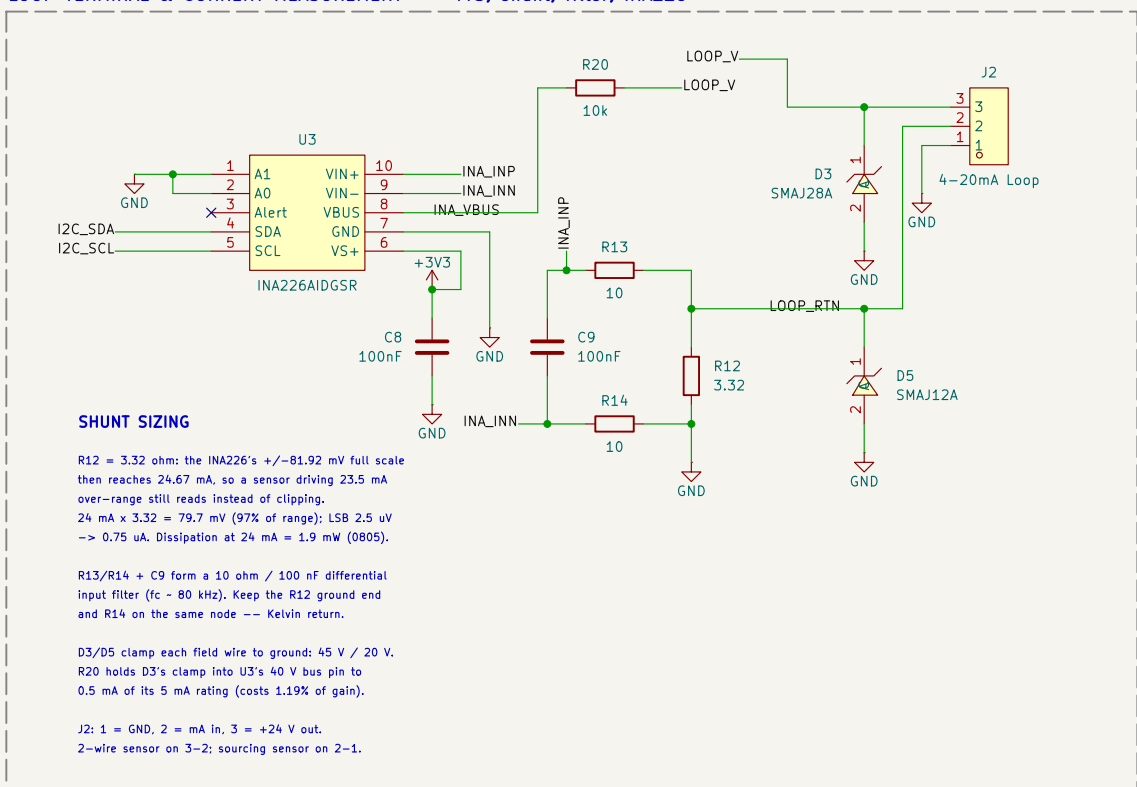
24 V BOOST CONVERTER - soft-start load switch, MT3608, feedback



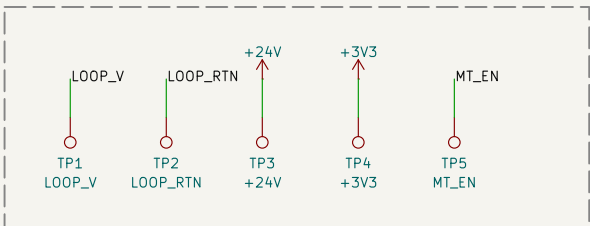
LOOP CURRENT LIMITER - 35 mA constant current, reverse blocking



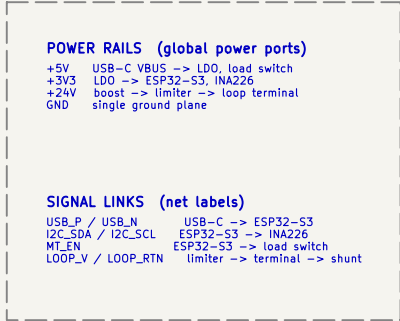
LOOP TERMINAL & CURRENT MEASUREMENT - TVS, shunt, filter, INA226



TEST POINTS



POWER RAILS & SIGNAL LINKS



USB-C powered 4-20 mA loop reader with onboard 24 V loop supply

Sheet: /
File: esp32-4to20ma-board.kicad_sch

Title: ESP32 4-20 mA ESPHome Sensor Board

Size: A2 Date: 2026-09-20

KiCad E.D.A. 10.0.6

Rev: G

Id: 1/1